

Swastika Khatua

Computer Science Engineering Student



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Summary

Computer Science Engineering student with strong fundamentals in machine learning and generative AI. Focused on agentic AI, with understanding of Retrieval-Augmented Generation (RAG), multi-step reasoning, and LLM orchestration, and hands-on experience building end-to-end ML and GenAI pipelines using Python, LangChain, vector databases, and FastAPI.

Education

- **Bachelor of Technology (BTech) in Computer Science and Engineering** Aug 2023 – Present
KIIT University CGPA: 9.17 / 10
Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Object-Oriented Programming
- **Holy Cross School** 2021–2023
ISC (Class XII): 94.75% 2020
ICSE (Class X): 98.6%

Skills

- Python
- Java
- C (Basics)
- Data Structures & Algorithms
- Object-Oriented Programming
- MySQL
- Machine Learning
- Deep Learning
- Natural Language Processing (NLP)
- Ollama
- RAG
- Scikit-learn
- TensorFlow (Keras)
- PyTorch
- LangGraph
- Streamlit
- Docker
- FastAPI
- Git & GitHub
- LangChain
- GenAI

Projects

- **Hybrid Movie Recommender System (ML – NLP)** 🔄
Built a hybrid recommendation system combining **content-based NLP techniques** (TF-IDF, cosine similarity) with **collaborative filtering** to generate personalized movie recommendations. Deployed model inference using **FastAPI REST APIs**.
- **Retrieval Augmented Generation (RAG) Research Paper Summarizer (GenAI – NLP)** 🔄
Designed and implemented a **Retrieval-Augmented Generation (RAG)** pipeline using **LangChain, ChromaDB,** and **Google Gemini** to generate context-grounded summaries from documents. Leveraged **semantic embeddings** and vector similarity search to reduce hallucinations and improve factual accuracy.
- **AI Financial Advisor (Agent-Based System)** 🔄
Built a multi-agent financial advisory platform using FastAPI with modular controllers, services, and agents. Implemented financial analysis, portfolio optimization, and simulation workflows with SQLite persistence. Deployed on Render + Streamlit for real-time advisory and chat-based interactions.

Certifications

- **Introduction to Deep Learning & Neural Networks with Keras** (IBM – Coursera, Jan 2026) 🔗
- **Oracle Cloud Infrastructure 2023 Certified Foundations Associate** (Oct 2025 – Oct 2027) 🔗
- **Oracle Cloud Infrastructure 2023 Certified AI Foundations Associate** (Oct 2025 – Oct 2027) 🔗
- **Oracle Cloud Infrastructure 2023 Certified Generative AI Professional** (Oct 2025 – Oct 2027) 🔗

Experience

- **Digital Marketing Intern, Veedol** (15 May 2025 – 28 June 2025) — Conducted **data-driven analysis** of marketing campaigns, improving engagement by 15%.
- **Grove** 🔗
Built an application featuring a **multi-node LLM pipeline** with **real-time web search, multi-provider AI routing** (Groq, OpenAI, Gemini), **JWT-based authentication**, and automated **knowledge crystallization** for persistent knowledge management.

Personal Skills

- Analytical problem-solving, teamwork, communication, and adaptability
- Languages: English (Professional), Hindi (Professional), Bengali (Native)